

Figure 7: **Left:** An example image-question pair from the VQS dataset and the corresponding execution trace of NS-CL. **Right:** Results on the VQS test set. Our model achieves a comparable results with the baselines.

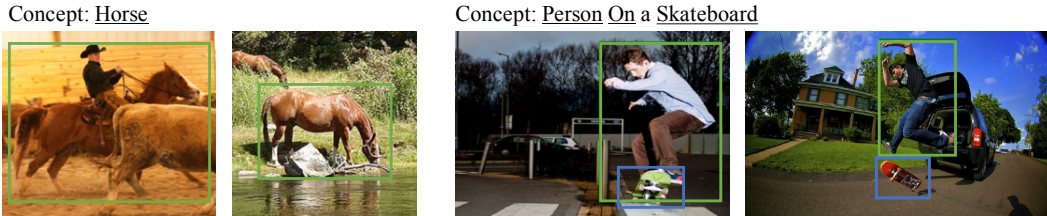


Figure 8: Concepts learned from VQS, including object categories, attributes, and relations.

shows competitive performance on QA accuracy, comparable with the MLP baseline (Jabri et al., 2016) and the MAC network (Hudson & Manning, 2018). Additional illustrative execution traces of NS-CL are in Appendix H. Beyond answering questions, NS-CL effectively learns visual concepts from data. Figure 8 shows examples of the learned visual concepts, including object categories, attributes, and relations. Experiment setup and implementation details are in Appendix G.2.

In this paper, we focus on a neuro-symbolic framework that learns visual concepts about object properties and relations. Indeed, visual question answering requires AI systems to reason about more general concepts such as events or activities (Levin, 1993). We leave the extension of NS-CL along this direction and its application to general VQA datasets (Antol et al., 2015) as future work.

5 DISCUSSION AND FUTURE WORK

We presented a method that jointly learns visual concepts, words, and semantic parsing of sentences from natural supervision. The proposed framework, NS-CL, learns by looking at images and reading paired questions and answers, without any explicit supervision such as class labels for objects. Our model learns visual concepts with remarkable accuracy. Based upon the learned concepts, our model achieves good results on question answering, and more importantly, generalizes well to new visual compositions, new visual concepts, and new domain specific languages.

The design of NS-CL suggests multiple research directions. First, constructing 3D object-based representations for realistic scenes needs further exploration (Anderson et al., 2018; Baradel et al., 2018). Second, our model assumes a domain-specific language for describing formal semantics. The integration of formal semantics into the processing of complex natural language would be meaningful future work (Artzi & Zettlemoyer, 2013; Oh et al., 2017). We hope our paper could motivate future research in visual concept learning, language learning, and compositionality.

Our framework can also be extended to other domains such as video understanding and robotic manipulation. Here, we would need to discover semantic representations for actions and interactions (e.g., push) beyond static spatial relations. Along this direction, researchers have studied building symbolic representations for skills (Konidaris et al., 2018) and learning instruction semantics from interaction (Oh et al., 2017) in constrained setups. Applying neuro-symbolic learning frameworks for concepts and skills would be meaningful future work toward robotic learning in complex interactive environments.

Acknowledgements. We thank Kexin Yi, Haoyue Shi, and Jon Gauthier for helpful discussions and suggestions. This work was supported in part by the Center for Brains, Minds and Machines (NSF STC award CCF-1231216), ONR MURI N00014-16-1-2007, MIT-IBM Watson AI Lab, and Facebook.

REFERENCES

- Omri Abend, Tom Kwiatkowski, Nathaniel J Smith, Sharon Goldwater, and Mark Steedman. Bootstrapping language acquisition. *Cognition*, 2017.
- Peter Anderson, Xiaodong He, Chris Buehler, Damien Teney, Mark Johnson, Stephen Gould, and Lei Zhang. Bottom-up and top-down attention for image captioning and visual question answering. In *CVPR*, 2018.
- Jacob Andreas, Marcus Rohrbach, Trevor Darrell, and Dan Klein. Learning to compose neural networks for question answering. In *NAACL-HLT*, 2016.
- Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. VQA: Visual question answering. In *ICCV*, 2015.
- Yoav Artzi and Luke Zettlemoyer. Weakly supervised learning of semantic parsers for mapping instructions to actions. *TACL*, 2013.
- Fabien Baradel, Natalia Neverova, Christian Wolf, Julien Mille, and Greg Mori. Object level visual reasoning in videos. In *ECCV*, 2018.
- Kan Chen, Jiang Wang, Liang-Chieh Chen, Haoyuan Gao, Wei Xu, and Ram Nevatia. Abc-cnn: An attention based convolutional neural network for visual question answering. *arXiv:1511.05960*, 2015.
- Kyunghyun Cho, Bart Van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. In *EMNLP*, 2014.
- Grzegorz Chrupała, Akos Kádár, and Afra Alishahi. Learning language through pictures. In *ACL*, 2015.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009.
- Jeffrey Donahue, Lisa Anne Hendricks, Sergio Guadarrama, Marcus Rohrbach, Subhashini Venugopalan, Kate Saenko, and Trevor Darrell. Long-term recurrent convolutional networks for visual recognition and description. In *CVPR*, 2015.
- Li Dong and Mirella Lapata. Language to logical form with neural attention. In *ACL*, 2016.
- Afsaneh Fazly, Afra Alishahi, and Suzanne Stevenson. A probabilistic computational model of cross-situational word learning. *Cognit. Sci.*, 2010.
- Chuang Gan, Yandong Li, Haoxiang Li, Chen Sun, and Boqing Gong. VQS: Linking segmentations to questions and answers for supervised attention in vqa and question-focused semantic segmentation. In *ICCV*, 2017.
- Siddha Ganju, Olga Russakovsky, and Abhinav Gupta. What’s in a question: Using visual questions as a form of supervision. In *CVPR*, 2017.
- Jon Gauthier, Roger Levy, and Joshua B Tenenbaum. Word learning and the acquisition of syntactic–semantic overhypotheses. In *CogSci*, 2018.
- Jonas Gehring, Michael Auli, David Grangier, Denis Yarats, and Yann N Dauphin. Convolutional sequence to sequence learning. In *ICML*, 2017.

- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2015.
- Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *ICCV*, 2017.
- Irina Higgins, Nicolas Sonnerat, Loic Matthey, Arka Pal, Christopher P Burgess, Matthew Botvinick, Demis Hassabis, and Alexander Lerchner. Scan: learning abstract hierarchical compositional visual concepts. In *ICLR*, 2018.
- Ronghang Hu, Jacob Andreas, Trevor Darrell, and Kate Saenko. Explainable neural computation via stack neural module networks. In *ECCV*, 2018.
- Drew A Hudson and Christopher D Manning. Compositional attention networks for machine reasoning. In *ICLR*, 2018.
- Allan Jabri, Armand Joulin, and Laurens van der Maaten. Revisiting visual question answering baselines. In *ECCV*, 2016.
- Justin Johnson, Ranjay Krishna, Michael Stark, Li-Jia Li, David Shamma, Michael Bernstein, and Li Fei-Fei. Image retrieval using scene graphs. In *CVPR*, 2015.
- Justin Johnson, Andrej Karpathy, and Li Fei-Fei. DenseCap: Fully convolutional localization networks for dense captioning. In *CVPR*, 2016.
- Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In *CVPR*, 2017a.
- Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Judy Hoffman, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Inferring and executing programs for visual reasoning. In *ICCV*, 2017b.
- Ryan Kiros, Ruslan Salakhutdinov, and Richard S Zemel. Unifying visual-semantic embeddings with multimodal neural language models. *arXiv:1411.2539*, 2014.
- George Konidaris, Leslie Pack Kaelbling, and Tomas Lozano-Perez. From skills to symbols: Learning symbolic representations for abstract high-level planning. *JAIR*, 2018.
- Tejas D Kulkarni, William F Whitney, Pushmeet Kohli, and Josh Tenenbaum. Deep convolutional inverse graphics network. In *NeurIPS*, 2015.
- Beth Levin. *English verb classes and alternations*, volume 1. University of Chicago Press, 1993.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *ECCV*, 2014.
- M Malinowski and M Fritz. A multi-world approach to question answering about real-world scenes based on uncertain input. In *NeurIPS*, 2014.
- Junhua Mao, Jiajing Xu, Kevin Jing, and Alan L Yuille. Training and evaluating multimodal word embeddings with large-scale web annotated images. In *NeurIPS*, 2016.
- David Mascharka, Philip Tran, Ryan Soklaski, and Arjun Majumdar. Transparency by design: Closing the gap between performance and interpretability in visual reasoning. In *CVPR*, 2018.
- Junhyuk Oh, Satinder Singh, Honglak Lee, and Pushmeet Kohli. Zero-shot task generalization with multi-task deep reinforcement learning. In *ICML*, 2017.
- Jeffrey Pennington, Richard Socher, and Christopher Manning. Glove: Global vectors for word representation. In *EMNLP*, 2014.
- Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *AAAI*, 2018.